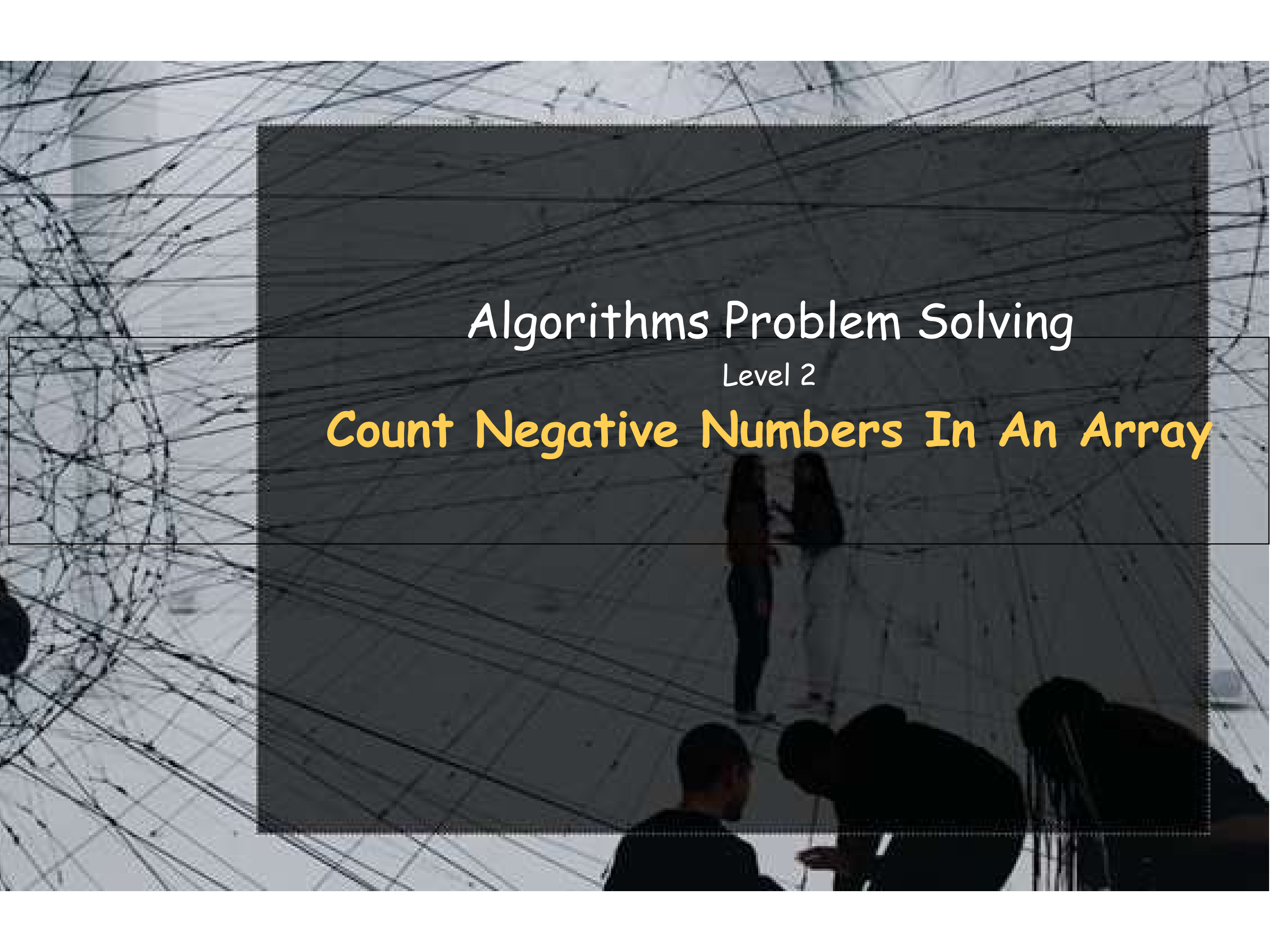


Algorithms Problem Solving

Level 2

Count Negative Numbers In An Array

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower portion of the image, the silhouettes of several people are visible, some standing and others sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

Algorithms Problem Solving

Level 2

Count Negative Numbers In An Array

Definition

- ◆ This program fills an array with random numbers from ``-100`` to ``100``.
- ◆ It counts how many negative numbers exist in the array.
- ◆ Finally, it prints the array elements and the total count of negative numbers.

Explanation

- ◆ `FillArrayWithRandomNumber()` fills the array with random numbers between `-100` and `100`.
- ◆ `CountNegativeNumbers()` loops through the array and checks each number.
- ◆ If the number is less than `0`, `count` is increased by `1`.
- ◆ The condition used in the code is:

```
if (arr[i] < 0)
    count++;
```
- ◆ Positive numbers and `0` are not counted.
- ◆ For example:
 - 16 → Negative → count = 1
 - 66 → Negative → count = 2
 - 77 → Positive → ignored
 - 13 → Positive → ignored
 - 10 → Negative → count = 3

Steps in main()

1. Ask the user for the number of elements.
2. Fill the array with random numbers from `-100` to `100`.
3. Print all array elements.
4. Loop through the array.
5. Check whether each number is less than `0`.
6. Increase `count` when the number is negative.
7. Return the total count.
8. Print the number of negative numbers.

💡 Example

◆ `Array Elements` :

22 94 -16 -66 77 13 -10 -88 -89 -5

◆ Negative numbers are:

-16 -66 -10 -88 -89 -5

◆ There are `6` negative numbers.

➡ Result

Array Elements: 22 94 -16 -66 77 13 -10 -88 -89 -5

Negative Numbers count is : 6

Note

◆ Important statement:

```
if (arr[i] < 0)
```

```
    count++;
```

◆ Only numbers smaller than `0` are counted.

◆ `0` is not negative, so it is ignored.

◆ `count` starts from `0` and increases for every negative number.

◆ `ReadArrayElements()` and `AddElementToArray()` are defined but not used in this program.

⊘ Warning

- ◆ `arr[100]` can store a maximum of `100` elements, so `arrLength` should not be greater than `100`.
- ◆ `FillArrayWithRandomNumber()` does not validate the entered length.
- ◆ `cmath` and `corecrt_math_defines.h` are not needed.
- ◆ `rand()` and `time()` should ideally have their required headers included explicitly.
- ◆ `rand()` generates pseudo-random numbers, not security-grade random numbers.



Comparison

▣ `CountNegativeNumbers()`

- ✓ Counts numbers less than `0`.
- ✓ Ignores positive numbers and `0`.
- ✓ Returns the total number of negative elements.

▤ `CountPositiveNumbers()`

- ✓ Counts numbers greater than `0` when using the correct positive condition.
- ✓ Ignores negative numbers and `0`.
- ✓ Returns the total number of positive elements.

✦ Summary

- ✓ Fill the array with random numbers from `-100` to `100`.
- ✓ Check every element.
- ✓ Count numbers that are less than `0`.
- ✓ Ignore positive numbers and `0`.
- ✓ Print the total number of negative elements.

Thank you 😊